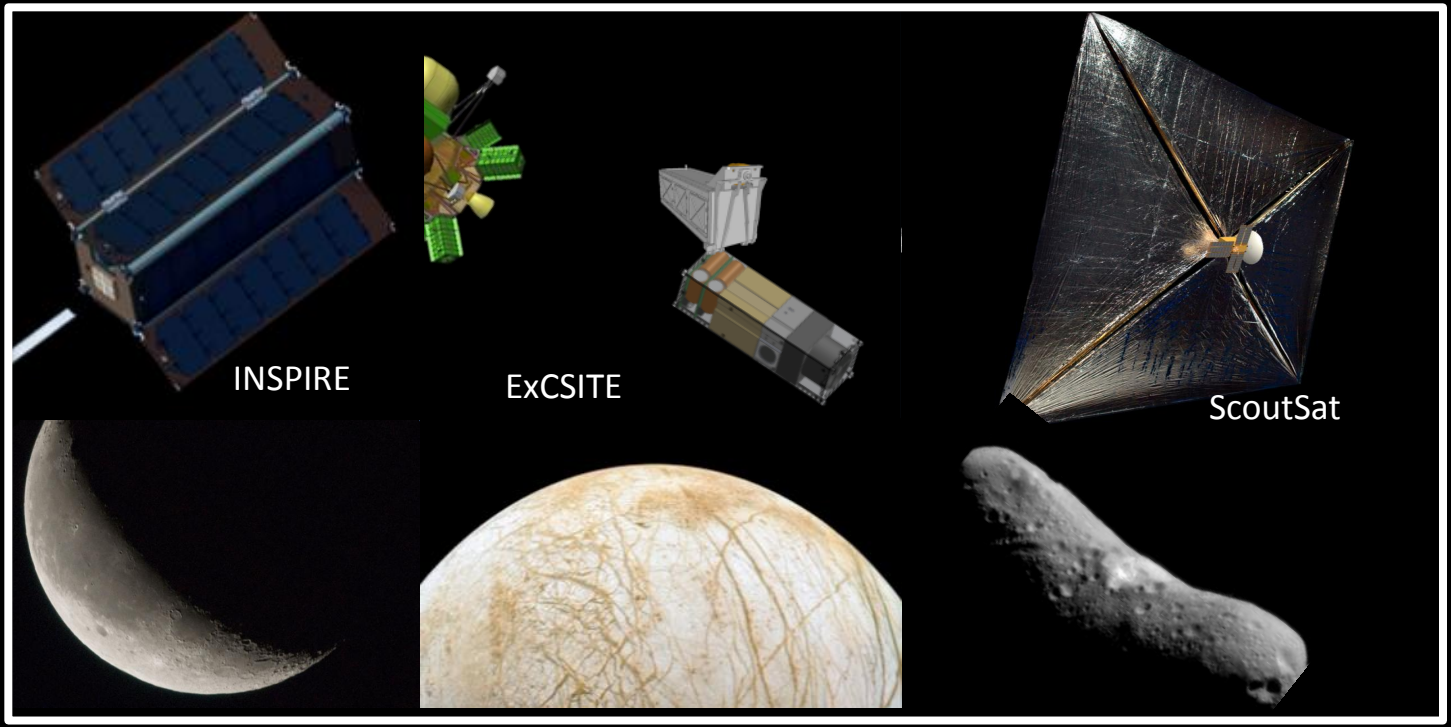
Next Generation SmallSat - Dare to Explore Where No Craft Has Gone Before

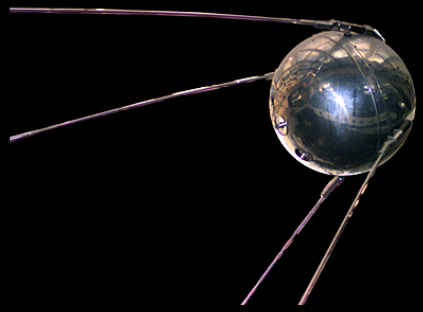


Julie Castillo-Rogez

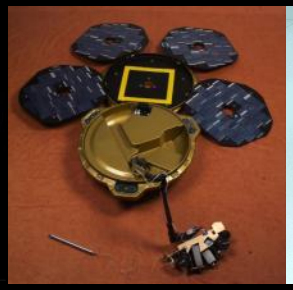
Andy Klesh, Sara Spangelo (JPL)

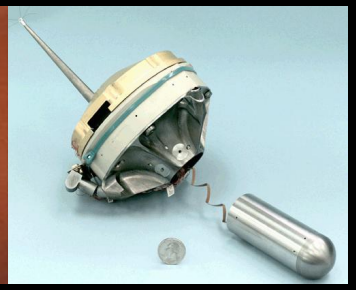
Classes of SmallSats

MiniSats (100 – 500 kg)



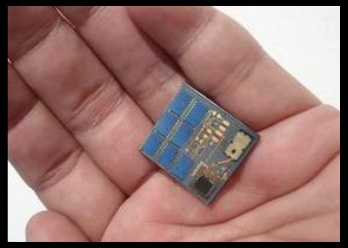
MicroSats (10 – 100 kg)





NanoSats (1 – 10 kg)

PicoSats (0.1 – 1 kg)



Molecular (Mole) Sats

(10^{-6} – 10^{-2} kg) ???

Novel Science Enabled by Nano-Spacecraft

- Smart instruments
- Access to unique vantage points
- Multi-site exploration at low cost
- Fractionated architectures
  - Distributed measurements
  - Fractionated payload
- Cooperation between multiple assets
  - Complementary vantage points
  - Risk assessment by sacrificial probe
  - Mothership watches daughtership interaction with target

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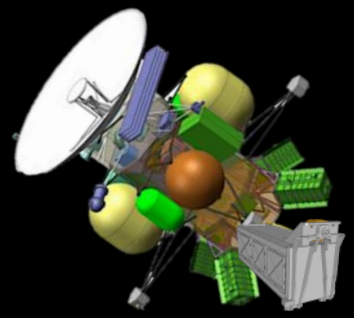
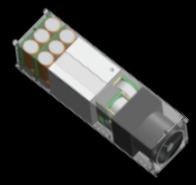
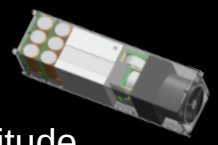
ExCSITE: Explorer Cubesat for Student Involvement in Travels to Europa

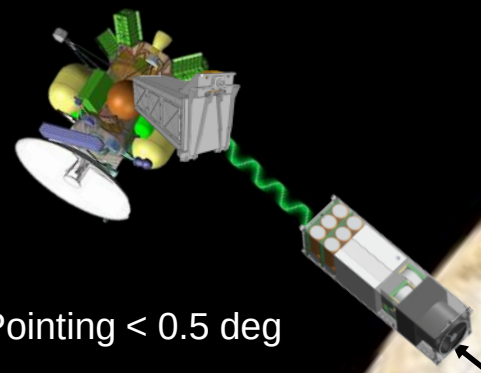
J. Castillo-Rogez, A. Frick, A. Klesh, J. Boland, P. Adell, H. Garrett, R. Staehle, N. Strange (JPL), D. Turner, V. Angelopoulos (UCLA), G. Collins (Wheaton College), J. Puig-Suari (Cal Poly)







Check altitude, re-orient



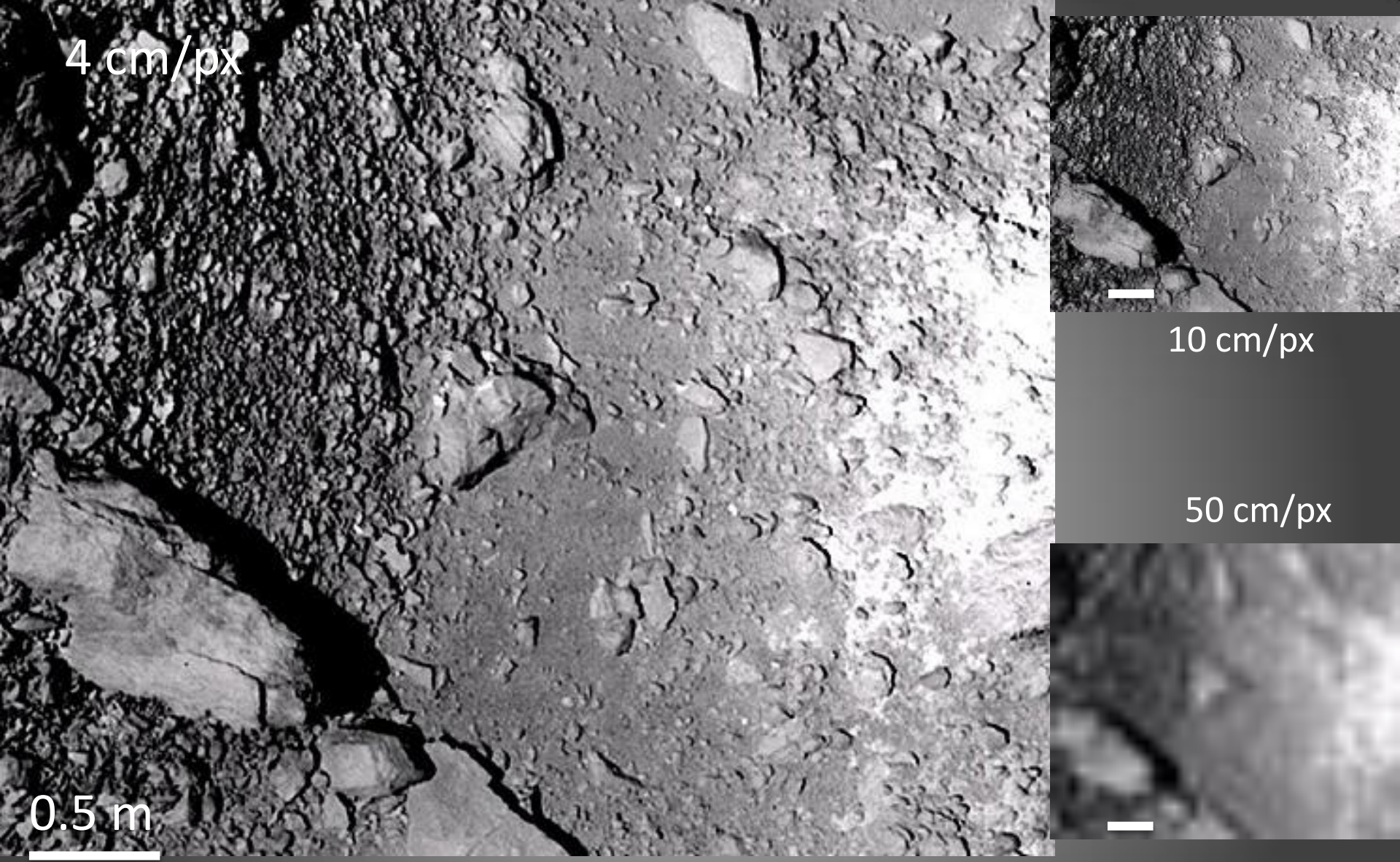
Pointing < 0.5 deg

Landing ellipse tbd (600x km^2)

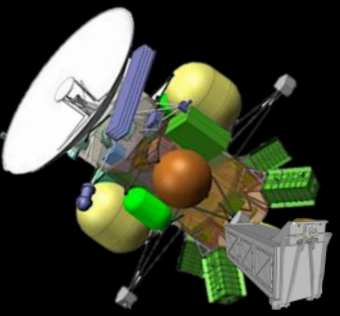
Impact angle > 70 deg,

Best resolution < 10 cm

Illumination direction 45-70 deg.

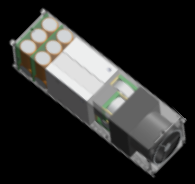


Goals: High-resolution imaging for in situ exploration reconnaissance, fields and particle mapping

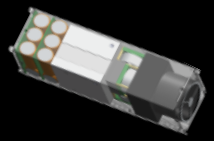


ΔV from deployer

Time/Distance
of deployment



Check altitude,
re-orient



ADCS
Intelligent C&DH?

Batteries

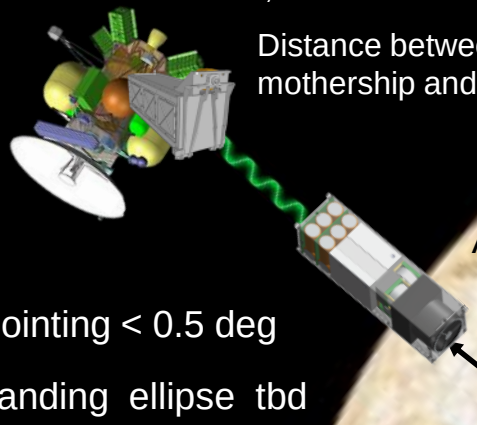
Compres
Factor

Deployed Instrument
Telecom frequency
Gain, Power

Distance between
mothership and Cubesat

Shielding strategy

Thermal strategy



Altitude of last image

Pointing < 0.5 deg

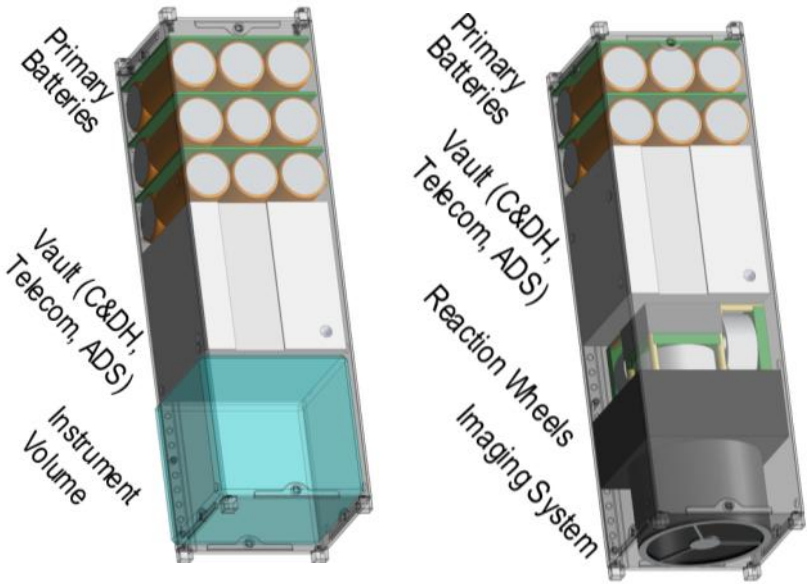
Best resolution < 10 c

Landing ellipse tbd
(600x km^2)

Illumination direction
45-70 deg.

Impact angle > 70 deg,

ExCSITE – Facing Europa Environment Challenges



- Radiations – sets the bar high
  (e.g., no star tracker solution)
- Limit impact on mothership
  (e.g., telecom)
- Planetary protection drives/limits systems choice
- Develop a multi-mission platform

→ Preserved Cubesat form factor
→Vault to protect electronics and instruments to be provided by universities and industrial partners
→ Smart packaging (some systems shield others)
→ Grant navigation control and telecom responsibility on deployer
→ And lots of cool features…

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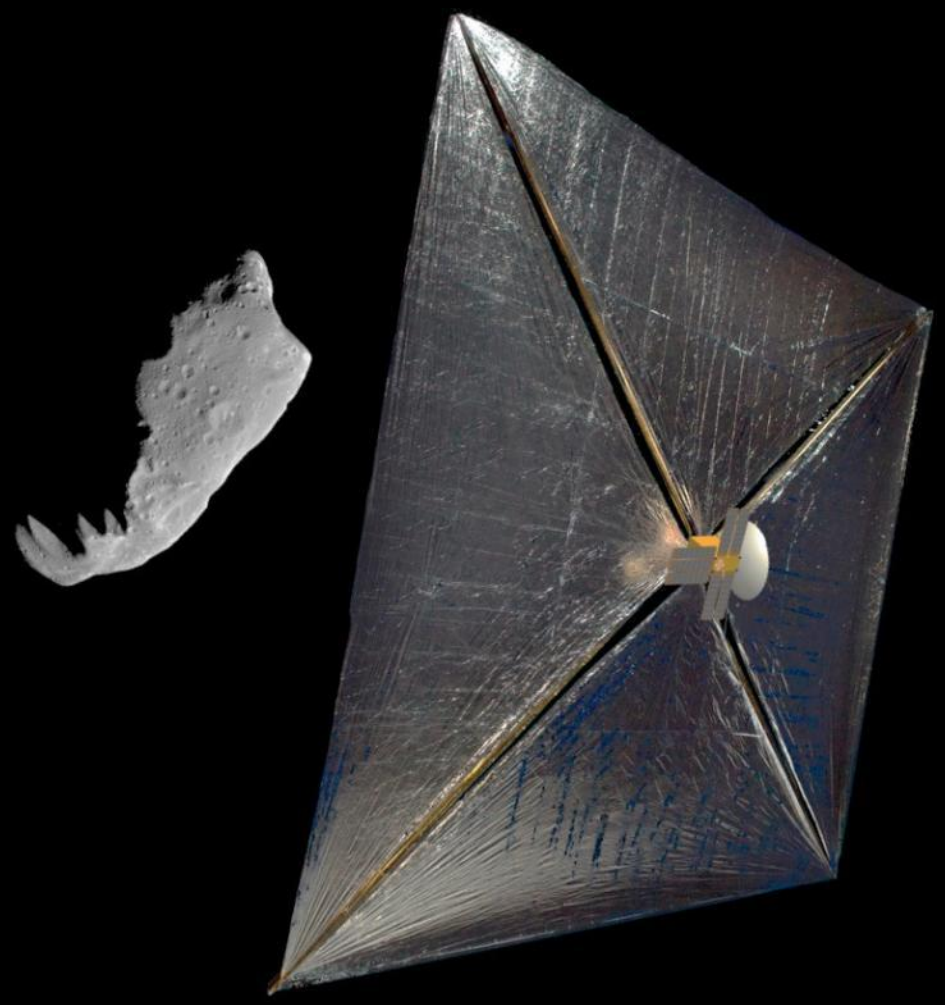
ScoutSat

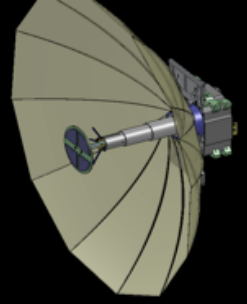


- **Goal**: Retire NEA knowledge gaps for human Exploration targets
- Perform 5 cm/px imaging
- Access human exploration target (< 50 m)
- Solar sail enables high navigation flexibility, and access to high inclination targets
  - CubeSat-scale sail maneuvering viable using: momentum wheels, or Center-of-mass shift, Momentum dump feasible using small articulating solar panels

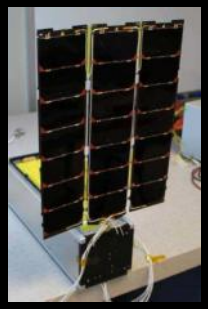
J. Castillo-Rogez, P. Kahn, P. Zarifian, A. Frick, P. Benazadeh, R. Staehle (JPL) P. Abell (JSC), U. Austin

ScoutSat: Technical Implementation

- Form Factor: 6U (10 x 20 x 30 cm^3)
  - 2U: Solar Sail and deployment mechanism
  - 1.5U: Instrument
  - 1.5U: ADACS, C&DH, Power, other 1U: Telecom
- Estimated mass: 12 kg w/ subsystem level cont.
- Leveraging COTS and emerging SmallSat technologies, but requires screeing for reliability
- Capabilities:
  - Generate 37 W at 1.2 $AU_{\odot}$
  - Store 30 Wh
  - Downlink 1 kbps at a AU_{+}
  - 0.003° pointing control (1σ), 2-axis
  - 0.007° pointing control (1σ), 1-axis
  - Slew rate of 10°/sec
  - Solar Sail area of 32 m^2
  - UT Austin cold gas thrusters
  - Passive thermal design

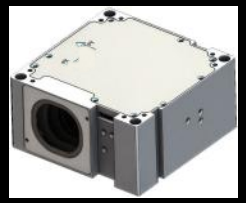


Boeing 0.5 m^2 deployable HGA



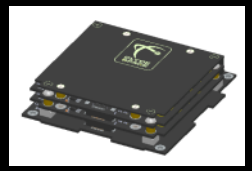
MMA 3x 10 x 30 cm^2 solar panels

COVE FPGA Board



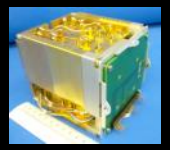
BCT XACT ADACS Module

Clyde Space EPS System

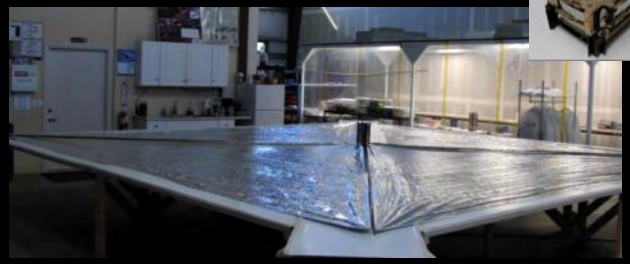


Clyde Space 30 Wh Battery

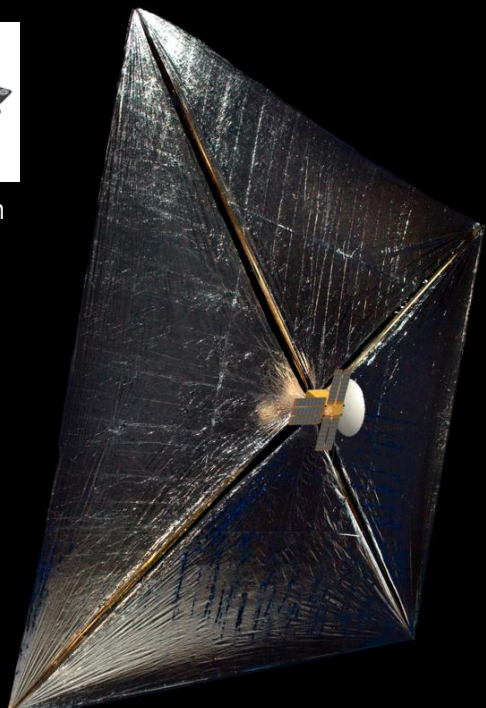
MARDI Camera



JPL X-Band IRIS Radio



Stellar Exploration's LightSail-1 solar sail



ScoutSat: Key Challenges

- Fine solar sail maneuvering remains to be demonstrated
  - Operational understanding of solar-sail navigation and attitude control without development
- Telecommunication (1kbps)

Summary - Technological Gaps

- Deployer and deployment strategy
  - Depending on perceived value, mothership may, or may not, change course
  - Limited GNC capability or lack of mature, flexible propulsion systems
  - Long cruise requires sturdier deployer for Cubesat (shielding, door lock, umbilical)
  - For controlled deployment ("smart" deployer)
- Telecomm (DTE, relay)
- Odometry/situational awareness
- Autonomous navigation
- Low power instruments
- More performant batteries/energy solutions
- Planetary protection strategy

A nice challenge, nothing out of reach

Way Forward

- Pursue the development of an **engineering standard** for deep space nano-spacecraft
- Need to increase heritage in mission design
  - In order to define multi-mission approaches to technological gap solving
  - Understand how science return scales with size
  - Quantify the trade between low-cost/simple operation nanosat and added complexity to mothership and operations
- Address the “risk” question
  - Where is the sweetspot between simplicity, cost, and risk?
  - Risk posture and redundancy
  - Mission insurance